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A

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SignVision: A Comprehensive Hybrid Model for Real-time Traffic and Safety Sign Detection

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DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgement has been made in the text.

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ABSTRACT

Sign recognition plays a vital role in many applications such as autonomous driving, driving assistance and industrial safety. While there is much research on traffic sign recognition, there has been little research on others such as industrial signs or safety signs that comply with the ISO 7010 standard which are needed for industrial safety. This project introduces SignVision, a comprehensive hybrid system to achieve fast and efficient detection and recognition of signs of different types including traffic signs from the German Traffic Sign Recognition Benchmark (GTSRB) and different safety warning signs from the ISO 7010 standard.

We employ a two stage hybrid system. In the first stage, we use traditional computer vision methods (color segmentation in the HSV color space, morphological operations and shape detection) to rapidly detect potential signs and reduce the search area. In the second stage we train a new, lightweight Convolutional Neural Network (CNN) classifier to accurately identify and classify the candidates into 53 sign classes.

Our two-stage approach addresses the "domain gap problem" where traditional methods (HOG+SVM) that perform very well in the traffic signs domain (95.8% accuracy on GTSRB) struggle with safety signs (71.2% accuracy on our SafetySign-10 dataset) due to their inability to discriminate complex pictograms. Our CNN based classifier achieves real-time (28 FPS) classification with 99.1% accuracy on GTSRB and 97.4% accuracy on SafetySign-10.

We validated the system in several experiments over two different and challenging datasets and three real-world scenarios of occlusion, low light and confusing classes. This demonstrates that the "classical + deep" approach is a practical and efficient solution that combines the speed of classical computer vision and the accuracy of deep learning techniques to achieve real-time and accurate traffic sign recognition across different sign domains.

Keywords: Sign Detection, Traffic Sign Recognition (TSR), Safety Signs, Computer Vision, Real-time Systems, Convolutional Neural Network (CNN), HSV Color Segmentation, Deep Learning, Hybrid Approach, GTSRB, ISO 7010.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
ADAS	Advanced Driver Assistance Systems
AGV	Autonomous Ground Vehicle
BGR	Blue Green Red (color space)
CNN	Convolutional Neural Network
CVPR	Conference on Computer Vision and Pattern Recognition
FPS	Frames Per Second
GPU	Graphics Processing Unit
GTSRB	German Traffic Sign Recognition Benchmark
HOG	Histogram of Oriented Gradients
HSV	Hue Saturation Value (color space)
ISO	International Organization for Standardization
mAP	Mean Average Precision
ReLU	Rectified Linear Unit
R-CNN	Region-based Convolutional Neural Network
ROI	Region of Interest
SIFT	Scale-Invariant Feature Transform
SS-10	SafetySign-10 Dataset
SURF	Speeded Up Robust Features
SVM	Support Vector Machine
TSR	Traffic Sign Recognition
YOLO	You Only Look Once
VRAM	Video Random Access Memory

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Signs are crucial to communicate non-verbal, strong information, regulations and warnings in complex public and industrial settings. Whether it be the speed limit signs on the motorway or the warning signs in the factory, signs represent a universal language which is independent of spoken words. Their timely and accurate detection by humans and machines is essential for safety, compliance and efficient functioning of transportation and industrial systems.

For autonomous vehicles and Advanced Driver Assistance Systems (ADAS) to function safely, legally and quickly, they must understand and comply with traffic signs [1]. Automatic detection and recognition of traffic signs is not just a desirable, but an essential feature of road safety. If the vehicle doesn't detect a stop sign or a speed zone change, it is dangerous for its passengers and other drivers. Likewise, signs are critical in industrial environments, like factories and construction sites, to prevent accidents, monitor regulations and aid in emergency escape routes [5]. Signs are important for human safety and safety of autonomous ground vehicles (AGVs) in industrial applications.

Traditionally, most work has been on Traffic Sign Recognition (TSR) [2]. Traffic signs are designed for use on roads, and the standard colours and shapes are defined in international agreements such as the Vienna Convention on Road Signs and Signals (1968) [7]. This makes traffic signs easy to detect: their colours (red, blue, yellow) are naturally different from the background, and their shapes (circle, triangle, octagon) are regular and well defined. The early approaches for traffic sign detection were successful in segmenting the required colour in an HSV or HSI color space [3], detecting the shape and classifying with hand-crafted features such as the Histogram of Oriented Gradients (HOG) [4], the Scale-Invariant Feature Transform (SIFT) [8] or the Speeded Up Robust Features (SURF) [9], and a Support Vector Machine (SVM) classifier. These methods were successful for the small number of classes and the mostly numeric or simple symbolic content of traffic signs.

But these conventional methods face inherent challenges with other types of signs. Safety signs, as defined in the ISO 7010 [5] standard, are much more challenging. They are often rectangular in shape, rather than circular or triangular, use different colours (yellow for warning, green for safe, blue for mandatory) and most importantly, the content of these signs consists of a variety of pictograms, rather than simple alphanumeric information. A HOG based classifier that was trained to detect a "50" sign would struggle to tell the difference between a "biohazard" symbol and a "radiation" symbol, despite the fact that these transmit fundamentally different safety messages.

Computer vision and object detection have been revolutionised by deep learning, specifically Convolutional Neural Networks (CNNs). Early research, such as VGGNet by Simonyan and Zisserman [14], demonstrated the success of very deep CNNs with small (3×3) convolution filters for state-of-the-art performance on large-scale image classification. The VGGNet architecture, which exclusively used small convolution filters, showed that increasing the depth of the network is crucial to achieve high accuracy, and that its features can be used for other tasks.

Meanwhile, approaches like YOLO (You Only Look Once) by Redmon et al. [13] framed object detection as a regression problem and obtained real-time detection (45 frames per second). YOLO's unified approach, which produces class probabilities and bounding boxes in a single evaluation, demonstrated that it's possible to achieve real-time and high-accuracy detections. However, end-to-end deep learning based object detectors like YOLO require large training data for each type of traffic sign to be detected and powerful computing resources for inference.

The German Traffic Sign Recognition Benchmark (GTSRB) [1] dataset is the standard for traffic sign recognition. The GTSRB dataset is a collection of more than 50,000 traffic signs of 43 different classes captured in real-world scenarios with varying illumination, weather conditions, occlusions and rotations. The benchmark has spurred a range of research innovations and the state-of-the-art systems are now on par with human performance.

In this project, we present SignVision, an integrated system for traffic sign detection and safety sign detection tasks. This hybrid system utilises efficient conventional computer vision techniques for candidate generation, and a powerful deep learning classifier for recognition, to provide a fast hybrid system that can deal with both tasks in a scalable manner.

1.2 Problem Statement

To detect signs in the wild, there are numerous challenges to be overcome:

- **Environmental Conditions:** Signs must be detected in a variety of lighting conditions - overexposure and glare from the sun, low light levels and low contrast during dusk and night. Precipitation (rain, fog and snow) can hide signs or distort them or produce reflections. Motion blur also occurs at high speed.
- **Signs are Not Perfect:** Signs are not perfect. They can be partially hidden by trees, cars or buildings. They can be faded and this is an issue if they are segmented on the basis of color. They can be obscured by dirt or graffiti, or even vandalised. Failing to detect a partially obscured stop sign can have disastrous consequences for autonomous cars.
- **Rotation and Distortion:** We normally do not see signs from the front. This skews, rotates and scales (further signs are smaller on the retina) the shape. Distortion and rotation must be taken into account in a detection algorithm.
- **The "Domain Gap" Problem:** It is solved by SignVision. One type of sign cannot be detected by models trained on the other type. This is due to the different conventions of the pictogram, colour and shape of these two types of signs. This has resulted in, until now, stand-alone traffic sign and safety sign detection systems.
- **Real-Time Detection:** For ADAS or active-safety systems at industrial sites, detection and classification needs to be performed in real-time (typically at a speed of >15 FPS) for timely actions. This rules out the possibility of using heavy deep learning models such as two-stage object detectors (such as Faster R-CNN [10]) unless one has very fast hardware.

These factors are not independent. A sign with low contrast, visibility and occlusion presents a complex problem that cannot be addressed with a single method. SignVision aims to address this issue with its hybrid approach.

1.3 Objectives

This project aims to:

- Build an effective hybrid sign detector (SignVision) using computer vision and deep learning methods to achieve efficient and accurate sign detection for different types of signs.
- Suggest a fast way to pick out colors to find road signs so it works reliably even if the background or lighting changes.

- Design a lightweight CNN architecture for accurate classification of 53 sign classes (43 traffic signs from GTSRB + 10 safety signs from ISO 7010) in a fast and efficient manner.
- Demonstrate the power of deep learning classification over other traditional HOG+SVM methods, particularly for complex pictograms of safety signs.
- Test the system using the experiments on the existing dataset (GTSRB) and a new dataset of safety signs (SafetySign-10) and difficult real-world conditions when the signs are obscured, lowly illuminated and share similar classes.
- Achieve real-time speed (≥ 15 FPS) for ADAS and safety systems in industry.

1.4 Scope of the Project

This project includes design, implementation and testing of the SignVision hybrid detection system. The scope includes: (i) the design of the traditional CV pipeline for sign candidate detection using HSV segmentation, morphological processing and shape matching; (ii) the design and training of a lightweight CNN for classification of 53 sign types; (iii) the integration of the two steps into a single real-time pipeline; (iv) experimentation with the GTSRB dataset and a custom dataset SafetySign-10; and (v) case study analysis of difficult detection scenes.

The project does not cover: (i) deployment on embedded platforms or edge devices such as Raspberry Pi, which would require quantisation and optimisation of the CNN model, extending the scope of the project; (ii) detection of other signs not in the database or from other countries; (iii) integration of multiple sensors including LiDAR or infrared cameras; and (iv) integration with vehicle control systems or industrial safety PLCs.

The system is made so that it can be extended. The CNN classifier is "plug and play" so new signs from other countries and/or new safety signs can be learned by retraining just the classifier without changing the detection model. This is an improvement on systems like YOLO [13], which require retraining the entire system for new signs.

CHAPTER 2

LITERATURE REVIEW

2.1 Traditional Sign Detection Methods

Before the dominance of deep learning, traffic sign detection relied on handcrafted features and multi-stage pipelines designed to exploit the predictable geometry and vibrant colors of road signage. These methods focused on computational efficiency, often prioritizing the rapid elimination of non-sign background data to meet real-time processing demands.

2.1.1 Color-based Segmentation

Since the early 2000s, traffic sign detection has been based on color segmentation. Gomez-Moreno et al. [3] implemented a successful traffic sign detection system using color segmentation in the HSV color space. The clever idea is that the HSV representation allows for the separation of chromatic (Hue) components from illumination (Value) components, which makes color-based detection more robust to variations in illumination than methods that use the RGB color space. The HSV color space uses three components: Hue (the dominant wavelength, in degrees from 0° to 360°), Saturation (the purity of the color, from 0 to 1) and Value (the brightness, from 0 to 1). OpenCV uses 0-180 for Hue, 0-255 for Saturation and Value. This representation enables specifying the colors of signs as ranges in the Hue dimension, with Saturation and Value ranges to remove all desaturated or dark pixels that are unlikely to be part of signs. Kardkovacs et al. [2] designed a real-time traffic sign recognition system that performed color segmentation as the initial step of a pipeline. They achieved real-time operation by quickly discarding most of the image based on color. The idea of using fast and approximate techniques to produce a set of candidate detections, followed by more accurate but slower techniques to classify the candidates, became the default approach for real-time sign detection.

2.1.2 Histogram of Oriented Gradients (HOG)

The Histogram of Oriented Gradients (HOG) descriptor proposed by Dalal and Triggs [4] for human detection became a very popular feature descriptor for sign recognition. HOG represents the distribution of orientation gradients in small regions of the image, forming a

vector that describes the object's appearance and shape. The vector is constructed by breaking the image into small spatial regions (cells), computing a histogram of gradient directions over each cell and normalising the histograms across a larger spatial neighborhood (blocks) for contrast invariance. HOG features were particularly successful for traffic sign recognition since traffic signs are mainly identified by their shapes and the basic symbols or digits they carry. The gradient representation is naturally sensitive to the shapes of circles, triangles and the strokes of numbers, and HOG features worked extremely well in classifying speed limit, directional, and prohibition signs. When paired with a linear SVM, HOG+SVM systems reported greater than 95% accuracy on traffic sign datasets [2]. But HOG also has a key weakness: it generates a global feature vector that represents the overall gradient distribution of the image, but not the semantic relationships between parts. This is a fatal weakness when classifying safety signs, where the main features are complex pictograms (trefoil radiation symbol or biohazard symbol) which need to be understood at a higher level.

2.1.3 SIFT and SURF Features

Another type of feature descriptors for sign recognition are Scale-Invariant Feature Transform (SIFT) by Lowe [8] and Speeded Up Robust Features (SURF) by Bay, Tuytelaars and Van Gool [9]. SIFT and SURF extract local features by first detecting local keypoints in the image and then calculating descriptors around these keypoints. These are scale invariant, rotation invariant and partially invariant to image distortion and illumination. Keypoint matching has been used for sign recognition using SIFT and SURF: a candidate sign is matched against a set of templates by comparing the keypoint descriptors. This method works well when the number of signs to be recognised is small and the signs are visually distinctive. But keypoint methods are slow compared to HOG, and are not recommended for real-time use. SURF is an alternative to SIFT, that is faster but with similar accuracy, but still slower than dense HOG extraction for real-time applications.

2.2 Deep Learning for Object Detection

Deep learning has fundamentally shifted the paradigm from manual feature engineering to automated representation learning. By utilizing neural architectures that mimic the hierarchical nature of human vision, these methods achieve unprecedented accuracy and robustness in detecting and classifying objects within complex environments.

2.2.1 Convolutional Neural Networks

Convolutional Neural Networks (CNNs) are a revolutionary approach in computer vision. Unlike traditional feature descriptors like HOG, CNNs learn features from data using backpropagation. Lower layers detect simple features like edges and textures, higher layers detect more complex parts and objects, and the deepest layers represent abstract semantics. LeCun et al. [6] pioneered the application of gradient-based learning to document classification with the LeNet architecture, showing that CNNs could be end-to-end trained for classification. LeNet's design, which alternates convolutional and pooling layers with fully connected layers, set the standard for all subsequent CNN designs. Ciresan, Meier and Schmidhuber [11] used multi-column deep neural networks for traffic sign recognition, setting a new state-of-the-art on the GTSRB dataset. They trained several CNNs on diverse preprocessing techniques on the input images, all of which were then averaged. This showed that deep learning could not only equal, but surpass human performance on traffic sign classification.

2.2.2 VGGNet Architecture

Karen Simonyan and Andrew Zisserman [14] were pioneers in their study of the impact of the depth of convolutional networks on accuracy in large-scale image classification. Their VGGNet architecture employed very small (3×3) filters for all of their convolutional layers, and showed that two successive 3×3 filters have an effective receptive field of 5×5 , and three 3×3 filters have an effective receptive field of 7×7 , but with fewer parameters and more non-linearity than simply using a single larger filter. The VGGNet models had between 11 weight layers (VGG-11) and 19 weight layers (VGG-19), with greater depth providing higher accuracy. The architecture laid the foundation for several CNN design principles: (i) use of small convolutional filters, (ii) depth is crucial for learning feature representations, and (iii) the number of filters to be doubled after every pooling layer ($64 \rightarrow 128 \rightarrow 256 \rightarrow 512$). These principles guided our SignVision CNN design. VGGNet won the ImageNet Large-Scale Visual Recognition Challenge (ILSVRC-2014) in the first and second places in the localisation and classification tracks, respectively. VGGNet representations were found to be generalisable to other datasets, showing good transfer learning performance. This is important for sign detection, as a range of sign types need to be recognised.

2.2.3 YOLO and Real-Time Detection

Redmon et al. [13] proposed YOLO (You Only Look Once), a different approach to object detection. Previous methods adapted classifiers for detection by running them at multiple locations and scales. YOLO looks at the entire image just once to instantly calculate exactly where objects are and what they are. YOLO splits the image into a $S \times S$ grid. Each cell makes predictions of B bounding boxes and confidence scores, and C conditional class probabilities. This monolithic approach is incredibly fast, with an inference speed of 45 frames per second (base model) and 155 frames per second (Fast YOLO). YOLO performs worse than slower two-stage detectors like Faster R-CNN [10] in terms of localisation error because it doesn't have a separate stage for generating candidate boxes, but it is less likely to predict false positives on background regions because it takes the entire image into account. Although YOLO and its variants (YOLOv2, YOLOv3, YOLOv4, YOLOv5, YOLOv6, YOLOv7 and YOLOv8) are the current state of the art in general object detection, they do have certain drawbacks for sign detection. First, they need lots of training data for training, and this may not be available for all signs. Second, they need to be fully retrained to incorporate new signs. Third, they require substantial GPU memory for real time detection. SignVision's hybrid model overcomes these challenges by separating the detection and classification tasks.

2.2.4 ResNet and Deep Residual Learning

He et al. [15] proposed deep residual learning with ResNet architecture to solve the issue of degradation in very deep networks. ResNet used skip connections (residual connections) to pass gradients directly through the network, allowing for the training of networks with more than 100 layers. ResNet achieved the best results on ImageNet with a 152-layer network, showing that very deep networks can be trained using residual connections.

The concept of residual learning has been adopted in the design of other CNNs, including for traffic sign recognition. But the high computational demands of very deep ResNet models can't be met by real-time sign detection algorithms running on conventional hardware without GPU support. Our SignVision CNN is based on deep feature learning but has a shallower structure tailored to the task of traffic sign recognition.

2.3 Traffic Sign Recognition Systems

2.3.1 The GTSRB Benchmark

The German Traffic Sign Recognition Benchmark (GTSRB), created by Stallkamp et al. [1], is the de-facto benchmark for traffic sign recognition algorithms. The dataset is derived from about 10 hours of video footage of traffic scenes on various road types in Germany during day-time. The set contains over 50,000 images of 43 traffic sign classes, with a natural imbalance of class frequencies as found on German roads.

The images in the GTSRB show high variability in appearance caused by distance (and hence resolution), changing lighting conditions (from sunlight to shadow), weather, partial occlusion, and skew. The benchmark employs a two-session test protocol in which participants classify the test images of two test sets of more than 12,500 images each. The GTSRB benchmark also reports the recognition accuracy for humans at 98.84%, serving as a useful reference for machine learning algorithms.

The GTSRB competition results show an evolution from classic to deep learning approaches. Initial submissions (HOG+SVM) performed at 95-96% recognition, whilst deep learning submissions steadily increased recognition accuracy, with the highest performing methods achieving more than 99% recognition (better than human performance). This trend justifies our use of deep learning for the classification step in SignVision.

2.3.2 Multi-column Deep Neural Networks

The first breakthrough on the GTSRB was achieved by Ciresan et al. [11] using a committee of deep neural networks. They used a multi-column deep neural network which learned several CNNs on differently preprocessed versions of the input and merged their results. This committee of CNNs achieved 99.46% accuracy on the GTSRB, far better than human performance. They showed that through the use of CNNs, data augmentation, and multiple CNNs, near-perfect classification accuracy could be achieved for sign recognition problems.

2.4 Safety Sign Detection

Unlike traffic sign recognition, there is comparatively little work on the topic of safety sign detection. The International Standardization Organization (ISO) 7010 [5] outlines a wide range of safety signs such as warning signs (yellow triangles), prohibition signs (red circles with diagonal bars), mandatory signs (blue circles) and safe condition signs (green rectangles). These play an important role in industrial workplaces, construction, laboratories and public buildings.

The main difficulty in detecting safety signs is the intricate nature and resemblance of the pictograms. While the information on traffic signs is often numerical (speed limit) or geometric (arrows, crosses) the information on safety signs is complex pictograms that represent a certain hazard or instruction. The biohazard sign, the radiation sign, the electrical hazard sign and the fire hazard sign are all yellow triangular signs with black pictograms, but they represent different safety messages. Misclassifying signs in an automated safety system could result in significant problems.

There is no large public benchmark dataset (such as GTSRB) for safety signs, which has slowed research in this domain. This inspired us to develop our SafetySign-10 (SS-10) dataset, which consists of more than 10,000 images of the 10 most frequent ISO 7010 safety signs found in university labs, offices and factories.

2.5 Comparative Analysis and Research Gap

Table 2.1: Comparison of Sign Detection Approaches

Approach	Method	Traffic Signs	Safety Signs	Real-time	Extensible
Gomez-Moreno [3]	HSV + Template	Good	Poor	Yes	No
Kardkovacs [2]	HSV + HOG + SVM	Good	Poor	Yes	Limited
Ciresan [11]	Multi-column DNN	Excellent	N/A	No	No
YOLO [13]	End-to-end CNN	Excellent	Possible	Yes	Requires full retrain
VGGNet [14]	Deep CNN classifier	Excellent	Good	No (heavy)	Transfer learning
SignVision (Ours)	HSV + Shape + CNN	Excellent	Excellent	Yes (28 FPS)	Pluggable CNN

The gaps in previous research that SignVision addresses is the lack of an integrated solution.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 System Overview

SignVision is a multi-stage hybrid computer vision approach, using both efficient classical algorithms and deep learning. The design philosophy is divide and conquer: fast but imprecise techniques are used to perform the very expensive operation of searching the entire image for candidate signs, while a deep but narrow CNN is used to perform the cognitively difficult task of deciding which exact sign is present from a few candidate signs.

Our approach has a number of advantages over end-to-end systems. First, the classical detection stage is run on the CPU, and can be run on the full frame in under 10 milliseconds, which means that more than 99% of the image can be rejected. Second, the CNN stage is only applied to a small number of potential detections (typically 2-5) per frame, instead of the entire image, and reduces the workload substantially for the GPU. Third, the detection and CNN stages can be updated independently, allowing to add new sign classes by simply retraining the CNN stage, without retraining the detection stage.

3.2 System Architecture and Workflow

The SignVision processing pipeline has five stages per frame:

Stage 1: Image Acquisition and Normalisation. The system captures a frame from the video source (webcam, video, image sequence) and standardises the size of the frame by resizing it to have a width of 640 pixels, keeping the aspect ratio. This facilitates the use of any image resolution and acceleration of the following steps in the pipeline.

Stage 2: Color-based Segmentation. The resized image is converted from the BGR to the HSV color space. Thresholding is then applied to produce masks for the four basic sign colours: red, blue, yellow and green. The four colour masks are then bitwise ORed to produce a mask of all the possible sign-coloured regions.

Stage 3: Candidate Detection. The merged mask is transformed (opened and closed) to remove noise and fill holes in signs. Contours are found to identify all regions. Geometric

filtering based on area, solidity, circularity and aspect ratio helps to remove non-sign regions, leaving a few candidate Regions of Interest (ROIs).

Stage 4: Classification. The candidate ROI is resized to 48 x 48 pixels, the input size of the CNN. A normalised ROI is fed to the CNN trained classifier, which produces a probability distribution over 53 sign types. The sign type is determined to be the class with the largest probability, if the confidence score passes a certain threshold.

Stage 5: Display. For each sign, the sign type, confidence score and the box coordinates are returned. This information can be overlaid on the image, saved or used in other systems such as vehicle control systems or factory safety monitoring.

3.3 Sign Detection Pipeline

The detection pipeline is the first stage of SignVision and is used to rapidly detect regions of the image that might contain signs. This pipeline is based on three techniques: colour segmentation, morphology and shape analysis.

Table 3.1: HSV Color Ranges for Sign Detection

Color	Hue Range (OpenCV)	Saturation Range	Value Range
Red (low)	0 – 10	70 – 255	50 – 255
Red (high)	170 – 180	70 – 255	50 – 255
Blue	100 – 140	70 – 255	50 – 255
Yellow	20 – 35	70 – 255	50 – 255
Green	40 – 90	70 – 255	50 – 255

The color ranges are tuned to cover the range of sign colors under different lighting conditions. We have two ranges for red because the red hue is at the end of the Hue spectrum in HSV. The Saturation minimum of 70 excludes pastel and washed-out colors that are unlikely to be signs. The Value minimum of 50 excludes very dark pixels that occur in deep shadows.

3.4 Sign Recognition and Classification

The second component of SignVision is the recognition stage, which aims to classify the candidate regions as one of the signs that we have interest in. We use a baseline of a HOG+SVM classifier, and compare it to our small CNN.

The HOG+SVM baseline approach computes a 9-bin HOG descriptor on the resized 48×48 candidate image, with 8×8 pixel cells and 2×2 cell blocks. The descriptor is classified by a linear SVM trained via "one-vs-rest" to handle multi-class classification. This baseline is in line with the state-of-the-art in traffic sign classification [2] and serves as a benchmark for the CNN approach. The CNN approach employs a lightweight, custom-designed architecture for the SignVision classification problem. This is detailed in Chapter 5. One of the main strengths of the CNN compared to HOG+SVM is the ability to learn semantic features that can separate visually similar but conceptually different signs - what's needed to classify safety signs.

CHAPTER 4

SIGN DETECTION METHODOLOGY

4.1 Color-based Segmentation

The first and most important step of the SignVision detection process is segmentation based on color. The reason for detecting signs via their colour is that signs are designed to catch the eye. The Vienna Convention on Road Signs and Signals [7] and ISO 7010 [5] standards prescribe particular colors for different types of signs. These mandated colours - red for prohibition and danger, blue for mandatory instructions, yellow for warning, and green for safe - are selected to stand out against natural and built environments.

The first step in our approach is to convert from BGR (Blue-Green-Red, OpenCV's default color space) to HSV (Hue-Saturation-Value) BGR encodes the perceived colour of an object as a mixture of colour and brightness: the BGR values of a red sign in shadow and a red sign in sunlight are quite different, even though to our eyes they are both "red". The HSV color space decouples these factors. The Hue dimension corresponds to the dominant wavelength (color), the Saturation dimension corresponds to the purity of the color and the Value dimension corresponds to brightness.

This is necessary for sign detection. By specifying the colors of signs as ranges in the Hue dimension, and using wide ranges in the Saturation and Value dimensions, a sign can be detected regardless of illumination. A red sign in sunny conditions will have a similar Hue value to a red sign in shadow, but a different Value value; using a wide range of Value values (50-255) will ensure both signs are segmented.

The segmentation results in a binary mask for each color. Red signs are segmented with two ranges of Hue values (0-10 and 170-180) since red is found on both sides of the Hue spectrum. These masks are then bitwise ORed together. The masks for the individual colors (red, blue, yellow, and green) are then combined into a combined mask via another bitwise OR operation. This mask identifies pixels in the image that are the four target colors.

The range of HSV values to use in the mask is a balance between recall (how many signs are detected) and precision (how many false positives are detected as signs due to similar colours). The Saturation minimum of 70 excludes pastel colours, which are seldom used for

signs but occur frequently in painted houses, clothes and vegetation. The minimum Value of 50 removes very dark pixels that are consistent with shadows where colors are often unreliable. These values were experimentally determined on a validation set of 500 images from a range of environments.

4.2 Morphological Operations

The binary mask obtained from color segmentation is bound to be noisy. Isolated pixels might be mistakenly detected due to natural or man-made objects that are within the desired color ranges. On the other hand, masks for signs may contain small holes, or gaps, due to reflections, shadows, or even damage to the signs themselves. These problems can be solved using morphological operations.

Morphological opening, which is an erosion followed by a dilation, erodes small, isolated areas (noise) and preserves the size and shape of larger areas. A 5×5 elliptical structuring element is used here. This removes false positives from small backgrounds such as car tail lights (which are in the red range), clothing and fruit or flowers.

Morphological closing (dilation followed by erosion), closes small gaps and holes in larger regions without changing their shape. This is essential for signs which have internal regions (the white stripes in a prohibition sign or the white number in a speed limit sign) which create holes in the color mask. The structuring element is a 7×7 elliptical kernel, a bit larger than the opening kernel to make sure that gaps are filled.

The sequence (opening then closing) is crucial. The initial opening operation clears out noise so that the closing operation does not inadvertently connect noise regions to form pseudo-sign candidates. Closing then connects regions within real signs, resulting in well-defined, connected binary regions as sign candidates.

The sizes of the kernels (5×5 for opening, 7×7 for closing) were selected by experimentation on the validation set. If these are too small, the image noise remains; if too large, the boundaries of small or distant signs are eroded. These sizes strike a good balance for the signs we have in the test data (which vary from 20×20 pixels for distant signs to 200×200 pixels for close signs).

4.3 Shape Analysis and ROI Extraction

Following the morphological operations, the noise-free binary mask contains connected components that are potential signs. Contour extraction, achieved via OpenCV's

findContours function (using the RETR_EXTERNAL retrieval mode, which only returns the external contours), identifies the contours of each connected region. The geometric properties of each contour are then tested to assess the likelihood of it being a sign.

4.3.1 Area Filtering

Contours with an area less than 500 pixels or greater than 25% of the total image area are immediately rejected. The lower area threshold filter removes very small areas that cannot possibly contain useful information about a sign, regardless of resolution. The maximum area threshold eliminates very large regions that cover the whole image, which often occur as sky or building facades with the same color as targeted signs. This first filter generally removes 60-70% of the detected contours, which greatly reduces the number of contours that need to be processed by more costly filters.

4.3.2 Shape Approximation

Each remaining contour is approximated with the Douglas-Peucker algorithm, using cv2.approxPolyDP with an epsilon of 2% of the arc length. It reduces the contour to its most important vertices, and the number of vertices can be used to determine the shape of the object. A shape with 3 vertices is a triangle (warning signs), 4 vertices is a rectangle/square (information signs, safety signs), 8 vertices is an octagon (stop signs) and more than 8 vertices is a potential circle (speed limit signs, prohibition signs).

The epsilon value determines the level of approximation. A lower epsilon will be more accurate but may generate more vertices than anticipated (noise). A greater epsilon results in a more simplified shape, but may not preserve the octagon of stop signs. The 2% value was empirically determined to work well for classification of our test cases with varying sign sizes and angles of view.

4.3.3 Solidity Test

The solidity of a shape is the area of its contour divided by the area of its convex hull. The solidity of a perfect solid shape (filled circle, triangle) is 1.0. Concave shapes or shapes with holes have a solidity less than 1.0. Our system requires a solidity above 0.85, which permits small imperfections in the sign's shape (caused by damage, occlusion or perspective) but rules out very irregular shapes that are unlikely to be signs.

4.3.4 Circularity Test

For shapes with more than 5 vertices, we calculate a shape's circularity, defined as $4\pi \times \text{Area} / \text{Perimeter}^2$. A circle has a circularity of 1.0. We set a minimum circularity of 0.7, to allow circles that are not at the front of the image (and will appear as ellipses), and circles with small amounts of damage or occlusion.

4.3.5 Aspect Ratio Test

For shapes with four vertices (possible rectangles or squares), the aspect ratio of the bounding rectangle is verified. Aspect ratios of traffic signs that are square should be close to 1.0; we accept aspect ratios in the range 0.7-1.3 to allow for perspective distortion. Safety signs are often rectangular, but are still accepted because the shape classification (rectangle or square) is valuable information for the CNN.

This two-stage geometric filtering process usually reduces the number of candidate regions from hundreds (after contour detection) to less than 5 per frame, with a selectivity rate greater than 99%. The remaining regions are potential sign regions, which are then fed to the CNN classifier.

CHAPTER 5

SIGN RECOGNITION AND CLASSIFICATION

5.1 Traditional Methods (HOG+SVM)

To compare with the proposed method, we use a pipeline of HOG and SVM. The Histogram of Oriented Gradients (HOG) feature descriptor, introduced by Dalal and Triggs [4] for detecting humans, is commonly used in traffic sign detection tasks [2]. The HOG descriptor is a distribution of oriented gradients in an image, which compactly represents the shape of the object.

Our HOG descriptor has parameters set as follows: cell size is 8×8 pixels, block size is 2×2 cells, the number of orientation bins is 9 (0° - 180°), and block normalisation is L2. This results in a 1,764-dimensional feature vector for a 48×48 -pixel input image. Our SVM classifier is linear with one-vs-rest multi-class, trained with scikit-learn's LinearSVC with $C=1.0$ and $\text{max_iter}=1000$.

On the traffic signs dataset (GTSRB), the HOG+SVM baseline reaches 95.8% accuracy, in line with published results [2]. This shows that traffic signs are well represented by HOG features, as they are mainly discriminated by their simple geometry (numbers, arrows, symbols).

In contrast, on our SafetySign-10 dataset, the HOG+SVM combination achieves only 71.2% accuracy. This substantial accuracy loss is a direct demonstration of the "domain gap". The HOG descriptor generates a global feature vector which represents the global gradient distribution but does not discriminate between safety sign pictograms which have similar global gradient distributions. For instance, biohazard and radiation warning signs are complex symbols in yellow triangles with a similar distribution of edges, making the SVM often confused.

5.2 Deep Learning Approach

We tackle the shortcomings of the HOG+SVM approach by using a Convolutional Neural Network (CNN) classifier instead. CNNs are more suited for complex classification than HOG because they learn hierarchical features. The first few convolutional layers learn to

recognise edges and textures; middle layers learn to recognise parts (such as the three "petals" of a radiation symbol or three "crescents" of a biohazard symbol); and later layers learn to recognise the full semantic meaning of each type of sign.

This feature learning capability is exactly what's needed to address the domain gap. Although HOG treats the biohazard and radiation symbols as similar distributions of gradient edge features, a CNN identifies the particular geometric shapes that separate them. The three-leaf clover of the radiation symbol and the three interlacing crescents of the biohazard symbol will activate different parts of the CNN, allowing them to be distinguished.

We use a CNN, rather than a conventional end-to-end object detector such as YOLO, for a good reason in the context of SignVision. The classical detection step has already identified and cropped potential regions of interest, so the CNN only needs to classify small cropped image patches, rather than the entire image. We are able to use a much shallower and quicker network than would be needed for detection in the entire image, enabling real-time performance without dedicated GPUs.

5.3 CNN Architecture Design

Our CNN design draws on LeNet-5 [6] and mini-VGG [14], tailored to the needs for sign classification: accurate classification of various pictograms, real-time performance for live operation, and low memory footprint for deployment.

Table 5.1: SignVision CNN Layer Configuration

Layer	Type	Output Shape	Parameters	Activation
Input	Image	48×48×3	0	—
Conv-1	Conv2D, 32 filters, 3×3	46×46×32	896	ReLU
BN-1	BatchNormalization	46×46×32	128	—
Pool-1	MaxPooling2D, 2×2	23×23×32	0	—
Conv-2	Conv2D, 64 filters, 3×3	21×21×64	18,496	ReLU
BN-2	BatchNormalization	21×21×64	256	—
Pool-2	MaxPooling2D, 2×2	10×10×64	0	—

Flatten	Flatten	6,400	0	—
Dense-1	Dense, 128 units	128	819,328	ReLU
Dropout	Dropout (0.5)	128	0	—
Dense-2	Dense, N units	N (53)	6,837	Softmax
Total			845,941	

The network is a simple two block convolutional network followed by a classifier. The convolutional blocks consist of convolution with ReLU activation, Batch Normalisation and Max Pooling. The classifier is made of a fully connected layer with ReLU activation, Dropout regularisation and Softmax output.

Design choices and considerations include:

- Input size $48 \times 48 \times 3$: This is large enough to distinguish between sign pictograms, but small enough to be quick. Having a larger input size of 64×64 provided a 0.3% improvement in accuracy with 40% longer inference time.
- Two convolutional layers: More layers (3-4 Conv layers) increased accuracy by less than 0.5% but slowed inference. Two layers are sufficient to learn features for sign recognition.
- Batch Normalisation: After each Conv layer to normalise the training, and get faster convergence and a small regularisation effect. Without BN, 2x longer training was required.
- Dropout 0.5: Applied before the output to prevent over-fitting. This large dropout is explored given the small training set and the need for good generalisation to unforeseen situations.
- Number of filters ($32 \rightarrow 64$): As in VGGNet [14] the number of filters is doubled after pooling. This enables to learn more complex features while reducing the number of parameters.

5.4 Training Methodology

The CNN is trained with the following settings:

- Optimiser: Adam with learning rate 0.001, $\beta_1 = 0.9$, $\beta_2 = 0.999$. Adam is a good optimiser with the advantages of momentum and adaptive learning rates for many loss functions.
- Loss function: Categorical cross-entropy, the typical loss for multi-class classification. The loss is $L = -\sum(y_i \cdot \log(\hat{y}_i))$, where y_i is the true label and $\hat{y}_i$ the prediction probability.
- Epochs: 50, with early stopping on validation accuracy with a patience of 10. Model typically converges after 30-35 epochs.

- Batch size: 64 images. This is a reasonable batch size that results in good gradient estimates, and can be stored in the 8GB VRAM of our NVIDIA GTX 1070 GPU.
 - Training data: 43 GTSRB classes (39,209 training images) and 10 SafetySign-10 classes (8,000 training images) - 47,209 training images in 53 classes.
- The training set includes a 80/20 train-validation split per class to allow for class balance in training. Early stopping and tuning of hyperparameters is performed on the training set. The test set (12,630 images from GTSRB, 2,000 from SS-10) is used for testing.

CHAPTER 6

IMPLEMENTATION DETAILS

6.1 Development Environment

Table 6.1: Development Environment Specifications

Component	Specification
Programming Language	Python 3.10
Classical CV Library	OpenCV 4.5 [12]
Numerical Computing	NumPy 1.21
Deep Learning Framework	TensorFlow 2.8 [16] with Keras API [17]
Machine Learning Utilities	Scikit-learn 1.0 (dataset splitting, HOG+SVM baseline)
GPU	NVIDIA GeForce GTX 1070, 8GB VRAM
CPU	Intel Core i7-8700K (6 cores, 12 threads, 3.7 GHz)
RAM	32 GB DDR4
Operating System	Ubuntu 20.04 LTS
CUDA Version	11.2
cuDNN Version	8.1

We chose a development framework for a balance between development convenience and inference time. There are many computer vision and machine learning libraries for Python 3.10. OpenCV 4.5 [12] provides efficient implementations of all traditional CV operations used in detection, including colour space conversions, morphological transforms and contours. TensorFlow 2.8 [16] with Keras API [17] provides an easy-to-use way to define, train and test the CNN classifier on the GPU.

The NVIDIA GTX 1070 GPU, while not the most powerful, is a mid-range GPU which can be expected to be accessible for deployment. If SignVision can be run in real-time on this GPU, it is likely to be faster on a more advanced GPU. The 8GB VRAM is sufficient for training (batch of up to 128 images) and inference (only one image at a time).

6.2 Dataset Preparation

Table 6.2: Dataset Statistics

Dataset	Classes	Training Images	Test Images	Source	Image Size
GTSRB [1]	43	39,209	12,630	Public benchmark	Variable (15×15 to 250×250)
SafetySign-10	10	8,000	2,000	Self-collected	Variable (50×50 to 400×400)
Combined	53	47,209	14,630	Merged	Resized to 48×48

6.2.1 GTSRB Dataset

SignVision is trained and evaluated on the German Traffic Sign Recognition Benchmark (GTSRB) [1]. The dataset is based on about 10 hours of video footage captured of traffic signs on German roads under various daytime conditions. It comprises over 50,000 images across 43 traffic sign classes, such as speed limit signs (8 classes), prohibition signs (10 classes), danger signs (12 classes) and mandatory signs (13 classes).

The images in GTSRB have considerable variability as found on real roads. The image sizes vary from 15×15 (small distant signs) to 250×250 (large close signs). Lighting conditions range from daylight to shadow. Signs are partially obscured by vegetation, other cars or poles. The distribution of classes is skewed, with some classes containing more than 2,000 images and others less than 200. The dataset is divided into a track-based split, with images from the same physical sign in either the training or test set, avoiding data leakage.

6.2.2 SafetySign-10 Dataset

Since there is no large public benchmark dataset for safety signs (like the GTSRB), we built the SafetySign-10 (SS-10) dataset. It includes over 10,000 images of the 10 most frequent ISO 7010 safety signs: Wet Floor, High Voltage, Biohazard, Radiation Warning, Eye Protection Required, No Entry, Fire Extinguisher, Emergency Exit, First Aid and Flammable Material.

We gathered images from various locations such as laboratories, offices, worksites and factories. The images were captured using the cameras of smartphones (iPhone 12 and Samsung Galaxy S21) from different distances, angles, and in different lighting conditions to introduce variability. Multiple shots were taken of each sign to provide varied training data. The photos were labelled and reviewed by two annotators.

The data was divided into training (8,000 images) and test (2,000 images) sets with a 80/20 ratio using stratified sampling. There are no duplicate images of the same physical sign in the training and test sets (the same approach used by the GTSRB).

6.3 Pipeline Optimization

The hybrid design is the major optimisation. By allowing the fast classical CV stage to discard over 99% of the image, the time-consuming CNN needs to ingest only a handful of candidate ROIs per image. This design optimisation is a key contributor to SignVision's superior speed compared to all-deep-learning approaches.

Additional optimisation techniques include:

- **Frame size:** The input frames are resized to 640 pixels width. This allows less pixels for colour segmentation and contouring, but enough for sign detection.
- **Colour masking:** The four colour masks are computed using NumPy's vectorised operations (`cv2.inRange`) function, which is applied to all pixels in parallel. Bitwise OR is used to compute the overall mask, avoiding pixel loops.
- **Contour hierarchy:** Contours are detected with `RETR_EXTERNAL`, returning only external contours, avoiding unnecessary work on smaller, internal contours.
- **Batching CNN inference:** If there are multiple candidates in a frame, they are in a single batch tensor and then fed to the CNN for inference, leveraging the GPU's parallel processing power.
- **Model optimisation:** The Keras model is compiled with TensorFlow XLA (Accelerated Linear Algebra) optimisation for quicker inference.

6.4 Data Augmentation Strategy

Augmenting the training data is crucial to learn a model that is able to cope with realistic variations when deployed in the wild. Our augmentation approach targets the environmental issues described in Chapter 1.

Table 6.3: Data Augmentation Parameters

Augmentation	Parameters	Purpose
Random Rotation	± 15 degrees	Simulate tilted signs and camera angle variations
Horizontal/Vertical Translation	Up to $\pm 10\%$ of image dimensions	Simulate imperfect centring of ROIs
Shearing	Up to ± 10 degrees	Simulate perspective distortion
Brightness Adjustment	$\pm 20\%$ of original brightness	Simulate lighting variations (sun, shadow, night)
Contrast Adjustment	$\pm 20\%$ of original contrast	Simulate atmospheric effects (fog, haze)
Random Horizontal Flip	50% probability	Augment training data (applied only to symmetric signs)
Gaussian Noise	$\sigma = 0.01$	Simulate sensor noise in low-light conditions

All augmentations are performed on-the-fly using TensorFlow's ImageDataGenerator, meaning that the data will be slightly different for each training epoch. This increases the training set size without taking up extra storage space. The augmentation parameters were selected to provide realistic changes without introducing unrealistic ones that might mislead the classifier.

CHAPTER 7

EXPERIMENTAL RESULTS AND ANALYSIS

7.1 Datasets Used

We tested SignVision on the GTSRB (43 traffic sign classes, 12,630 test images) and the SafetySign-10 (10 safety sign classes, 2,000 test images) datasets. The unified classifier was trained on all 53 classes, forcing the model to learn features to discriminate traffic signs from safety signs and between the different types of signs in each domain. This joint training is more complex than training two classifiers (one for traffic signs, one for safety signs) but shows SignVision's capacity to address the domain difference in one model.

7.2 Classification Accuracy

Table 7.1: Classification Accuracy Comparison

Classifier Model	GTSRB (Traffic)	SS-10 (Safety)	Combined (53 classes)	Inference Time
HOG + Linear SVM	95.8%	71.2%	89.3%	2.1 ms/image
HOG + RBF SVM	96.4%	73.8%	90.1%	3.8 ms/image
SignVision CNN (Ours)	99.1%	97.4%	98.7%	8.2 ms/image

These results show that CNNs outperform the HOG+SVM classifiers. On the GTSRB dataset, the CNN reaches 99.1% accuracy, whereas the HOG+SVM method reaches 95.8% accuracy, a substantial improvement that approaches human-level performance of 98.84%. More significantly, on the SafetySign-10 dataset, the CNN achieves 97.4% accuracy versus only 71.2% for HOG+SVM, a 26.2% improvement that confirms our suspicions about the limitations of HOG features for the complex pictograms of safety signs.

The performance of the HOG+SVM detector on safety signs is telling. We observe that the biggest confusion is among visually similar warning signs: biohazard/radiation (38% confusion rate), electrical hazard/flammable (25% confusion rate) and eye protection/general

mandatory (19% confusion rate). These signs have similar global shape and gradient distributions, to which HOG is sensitive. The CNN, which learns hierarchical features of each pictogram, can successfully discriminate these signs with a success rate of more than 97%.

Table 7.2: Per-Class Accuracy on GTSRB (Selected Classes)

Sign Class	HOG+SVM	CNN (Ours)	Improvement
Speed Limit 30	97.2%	99.5%	+2.3%
Speed Limit 50	96.8%	99.3%	+2.5%
Stop	99.1%	99.8%	+0.7%
No Entry	98.5%	99.6%	+1.1%
Yield (Triangle)	97.0%	99.4%	+2.4%
Pedestrian Crossing	94.2%	98.9%	+4.7%
Roundabout	93.5%	98.7%	+5.2%

7.3 Real-time Performance

Table 7.3: Real-time Performance Metrics

Metric	Value
Overall FPS	28 frames per second
Color Segmentation Time	3.2 ms per frame
Morphological Operations	1.8 ms per frame
Contour Detection + Filtering	2.4 ms per frame
CNN Inference (per ROI)	8.2 ms per ROI
Average ROIs per Frame	2.3
Total Pipeline Time	35.7 ms per frame
GPU Utilisation	42% (GTX 1070)
CPU Utilisation	28% (single core)

The system's frame rate is 28 FPS (tested on GTX 1070 + i7-8700K), well above the minimum 15 FPS required for ADAS. The benchmarking results show that the time spent on classical CV (color segmentation + morphological operations + filtering contours) accounts for 7.4 ms for each frame, while the CNN inference takes an average of 18.9 ms

per frame ($8.2 \text{ ms per ROI} \times 2.3 \text{ average ROIs}$). The rest is the time for image capture, resizing, and drawing the results.

The low CPU (14%) and GPU (42%) utilisation suggests that there is plenty of room for higher-resolution input, bigger batch processing, or even multiple CNN inference operations. This is crucial for real-world applications where the sign detection system needs to coexist with other perception tasks that use a GPU, such as lane detection, object tracking and depth estimation.

7.4 Case Studies

7.4.1 Case Study 1: Partial Occlusion

A Stop sign was partially hidden by a tree branch, on average 30% of the sign was not visible. The red areas of the sign were segmented by the colour segmenter. The shape analysis step had difficulty determining that the shape was octagonal, because it was partially obscured, but solidity and area filters were set to be quite tolerant (solidity > 0.85 , which means that up to 15% of the area can be outside the convex hull), so it was deemed to be a potential sign.

The ROI was cropped to cover 70% of the sign, but the CNN was able to identify it. The CNN was trained on data that was augmented with randomly occluded (up to 30% of the image) rectangular patches and translations, and it learned to recognise signs from partial information. The critical information that the CNN uses in order to recognise the sign - the letters "STO" and the white-on-red colour - were visible in the unoccluded area, so the sign was correctly classified with 89.3% confidence.

7.4.2 Case Study 2: Poor Illumination

A "Wet Floor" safety sign was in a dimly lit hallway. The value (V) of the HSV representation was very low ($V < 80$ in most parts of the sign), resulting in a difficult colour segmentation. We had a wide tolerance for the Value (V: 50-255) of our HSV threshold and this resulted in a noisy mask.

However, in this example, the closing operation was essential to join the yellow patches. This led to an oddly-shaped region, but the area and solidity filters retained it as a possible ROI. Our trained CNN (on brightness-adapted images with $\pm 20\%$ brightness change) could successfully classify this dark and grainy ROI as "Wet Floor" with an accuracy of 84.7%. A

HOG+SVM-based detector would fail in this case due to inaccurate gradient calculations due to low contrast and noise.

7.4.3 Case Study 3: High Inter-Class Similarity

This is an example of the domain gap and its solution, SignVision. We showed the system two signs of caution: "Biohazard" and "Radiation Warning". Both signs are yellow triangles with black symbols, and look similar. Our baseline (HOG+SVM) misclassified the biohazard sign as radiation in 38% of the test cases, showing the need for better descriptors than gradient-based features for these complex symbols.

Our CNN recognised both signs in all tests (100% for this pair). CNN activation visualisation (Grad-CAM) revealed that it is focusing on the different geometric features of the symbols: the three crescents of the biohazard symbol versus three-pronged trefoil and a central dot of the radiation symbol. These are the type of hierarchical structure CNNs learn but HOG descriptors do not.

Table 7.4: Case Study Results Summary

Case Study	Scenario	Detection Success	Classification Accuracy	Confidence
1: Partial Occlusion	Stop sign 30% occluded by tree	Yes	Correct (Stop)	89.3%
2: Poor Illumination	Wet Floor sign in dark hallway	Yes	Correct (Wet Floor)	84.7%
3: Inter-Class Similarity	Biohazard vs. Radiation	Yes	Both correct	96.1% / 97.8%

CHAPTER 8

ADVANTAGES, LIMITATIONS AND FUTURE SCOPE

8.1 Advantages

SignVision has several advantages over current sign detection systems:

- **Multi-Domain Sign Detection:** SignVision is the first sign detection system to detect and recognise both traffic and safety signs. This approach overcomes the domain gap between the two categories of signs, and removes the need to develop separate systems for each type of sign, thereby simplifying the deployment and reducing the cost.
- **Higher Accuracy on Safety Signs:** The CNN-based classifier achieves 97.4% accuracy on safety signs, improving over the HOG+SVM by 26.2% percentage points. This addresses the major gap of current systems on safety signs.
- **Efficiency in Real-Time:** SignVision runs at 28 FPS on a mid-level GPU (GTX 1070), and thus achieves real-time operation for ADAS and safety applications. The hybrid approach guarantees that the CNN only searches a small number of candidates per image, which decreases the CNN's processing time.
- **Illumination-Robustness:** The HSV-based color segmentation is naturally illumination-invariant, as is the HSV color model. Data augmentation during training also leads to invariance to illumination, contrast, rotation and occlusions.
- **Modularity:** The modular CNN architecture allows for the inclusion of new sign classes, through retraining of the CNN-based classifier. This is unlike end-to-end systems like YOLO [13] which require retraining.
- **Optimal Resource Usage:** Classical CV processing is done on the CPU, removing >99% of the image from the pipeline and thereby decreasing the GPU utilisation. The low GPU load of 42% can be used by other perception tasks in a multi-modal system.

Table 8.1: Comparison with State-of-the-Art Systems

System	Traffic Accuracy	Safety Accuracy	FPS	Multi-Domain	Extensible
HOG+SVM [2]	95.8%	71.2%	45+	No	No

YOLO v3 [13]	98.2%	N/A	30	Retrain needed	Full retrain
VGG-16 Classifier [14]	99.0%	95.1%	8	Yes	Transfer learning
Multi-column DNN [11]	99.5%	N/A	5	No	Full retrain
SignVision (Ours)	99.1%	97.4%	28	Yes	Pluggable CNN

8.2 Limitations

While SignVision has achieved good performance, there are some limitations that need to be overcome in future research:

- **Generalisation to Different Environments:** The system has yet to be tested in extreme weather environments like fog, snow and heavy rain. These can severely affect the colour-based segmentation by making colours less saturated and less visible. The V (Value) channel is significantly diminished in fog and rain, and could lead to a failure of the segmentation step.
- **Data Bias and Domain Adaptation:** The GTSRB includes only German traffic signs, which have different designs than other countries (e.g., American stop signs, Japanese road markings). The SafetySign-10 dataset is small and gathered from a narrow geographical area, and may be biased. The system may not perform well with new signs unless it's retrained with additional images.
- **Computational Efficiency:** While we claim our CNN is "lightweight", it still requires a GPU for real-time use. This prevents it from being run on low-powered edge devices such as Raspberry Pis or Arduinos, where an alternative method such as HOG+SVM may be the only viable solution, though less accurate. Quantisation and pruning of the model could resolve this but are yet to be investigated.
- **Cascaded System Failure Mode:** The system is only as good as the weakest link. If the color segmentation fails to detect a sign (e.g., a totally faded sign, a sign completely in shadow, or a sign of a different color), then the CNN classifier will fail to classify it. This is not a problem for end-to-end deep learning detectors such as YOLO, which directly learns features for detection from raw pixels.

- **Number of Sign Classes:** The system currently covers 53 sign classes (43 traffic signs + 10 safety signs). This covers the most common signs but for real-world applications, hundreds of different signs may be needed across different countries and industries. To support more classes, the CNN may need to be made deeper and more training data needed.
- **No Temporal Tracking:** The current approach does not feature temporal tracking. In video, past frame information (e.g. the known location and type of a detected sign) could be used to increase the confidence of the detection and decrease the number of false positives. The inclusion of a tracking component (Kalman filter, Hungarian algorithm) is an obvious next step.

8.3 Future Scope

Several promising directions for future work have been identified:

8.3.1 Domain Adaptation and Transfer Learning

The most promising future work is to use domain adaptation methods [18], which would allow the network to work with a new set of signs without having to train from scratch. Visual domain adaptation techniques can learn from a source domain (e.g., German traffic signs) to adapt to a target domain (e.g., American or Indian traffic signs) with limited data. This would greatly improve SignVision's geographic coverage.

Pre-training the CNN using ImageNet-trained models (like ResNet or EfficientNet) would also help to overcome limited data by transferring knowledge from a large dataset of images. This would especially help with rare sign types with limited data.

8.3.2 Model Compression for Edge Deployment

To allow for inference on low-power devices, model compression [19] methods such as pruning, quantisation and distillation should be investigated. Han, Mao and Dally [19] showed that deep neural networks can be compressed 10-50x with negligible loss in accuracy by jointly training a sparse network, quantising the weights to low bit-precision and Huffman coding the weights. If applied to our CNN, these techniques could allow real-time inference on devices like the NVIDIA Jetson Nano, or even smartphones.

8.3.3 Multi-Sensor Fusion

The system is vulnerable to extreme weather conditions and darkness. Adding other sensors, such as LiDAR (for distance and 3D shape) and infrared cameras (for darkness and poor

weather) would prevent this. LiDAR detection of flat objects at particular heights above the ground could offer an alternative candidate detector, which is not susceptible to color-based breakdowns.

8.3.4 Temporal Tracking and Video Analysis

To better detect signs in video, a tracking component could also be added that keeps track of the signs from one frame to the next, improving the robustness of detection, and introducing new possibilities such as counting signs, inferring distance to signs and estimating the vehicle's speed. A Kalman filter tracker could provide a prediction of where signs have been previously detected, minimising the need to search the entire frame and accelerating detection.

CHAPTER 9

CONCLUSION

In this project, we present SignVision, a hybrid, real-time system for traffic and safety sign detection. We found that the current state of research lacks a system that addresses both traffic and safety signs, and that classical computer vision approaches for traffic signs often struggle with the complex pictograms of safety signs.

We propose a hybrid approach that leverages efficient, classical computer vision techniques (HSV segmentation, morphological processing, and shape analysis) to detect candidate signs and a small, fast and accurate CNN classifier to learn and classify them. The classical part quickly filters out more than 99% of the image from consideration, enabling the CNN to devote its computational power to a relatively small number of high-quality candidates.

The results show the benefits of using deep learning for classification. On the GTSRB traffic sign dataset, SignVision achieves 99.1% accuracy - close to human performance. On our SafetySign-10 dataset, SignVision reaches 97.4% accuracy, a huge 26.2% improvement over the HOG+SVM baseline (71.2%). This gain supports our initial hypothesis that multi-level feature learning with CNNs is critical to recognise complex safety sign pictograms with similar global gradient structures.

Our three case studies also demonstrate the system's effectiveness against adversarial cases. SignVision accurately detects signs that are partially blocked (Case 1), signs in low light (Case 2) and signs that are similar and cause confusion for classical classifiers (Case 3). These examples show that by using the speed of classical computer vision to reduce the search space and the accuracy of deep learning to perform the final classification, SignVision achieves much better performance than either purely classical or purely deep approaches.

The major lesson from this research is that the "classical + deep" hybrid design is a very successful paradigm for real-time sign detection. By leveraging the efficiency of classical computer vision to narrow down the search space and the accuracy of deep learning to provide the final sign classification, SignVision achieves high accuracy and real-time operation (28 FPS) on modest computer hardware. The pluggable design allows for easy incorporation of additional sign classes.

The current system has several shortcomings, namely limited to fair weather conditions, requiring GPU hardware to achieve real-time performance, and having a narrow geographic scope in the training data. Our future research will address domain adaptation to other sign languages, model compression for deployment on the edge, sensor fusion for improved robustness in bad weather conditions, and temporal tracking in video streams.

Sign detection plays a key role in support of autonomous driving, advanced driver assistance systems (ADAS) and industrial safety automation. The increasing deployment of these systems will increase the need for robust and fast detection of a wide variety of signs. SignVision is an extensible, practical solution to this problem that shows how a unified system can be developed to bridge the gap between traffic and safety sign detection by drawing on a combination of classical and deep learning approaches.

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APPENDIX A

CNN ARCHITECTURE DETAILS

This appendix provides more technical details about the SignVision CNN architecture, including the rationale for each design decision and the other options that were evaluated during development.

A.1 Architecture Selection Process

The list of candidate architectures we made a choice between before settling on the final architecture includes: (1) LeNet-5 variant (2 Conv + 2 FC), (2) Mini-VGG (3 Conv + 2 FC), (3) MobileNet-V2 (pretrained, fine-tuned) and (4) ResNet-18 (pretrained, fine-tuned). Both were trained using the combined 53-class dataset at 50 epochs using the same hyperparameters.

Architecture	GTSRB Acc.	SS-10 Acc.	Parameters	Inference Time	FPS
LeNet-5 variant	97.2%	93.1%	312K	4.1 ms	52
Mini-VGG (selected)	99.1%	97.4%	846K	8.2 ms	28
MobileNet-V2	99.3%	97.8%	2.3M	14.5 ms	18
ResNet-18	99.4%	98.1%	11.2M	22.8 ms	12

The Mini-VGG architecture has been chosen as the optimal trade-off between accuracy and speed. MobileNet-V2 and ResNet-18 were only marginally more accurate (within the same range of +0.2-0.7) but much more expensive (1.8-2.8x slower inference). The LeNet-5 version was much faster, but with a 1.9-4.3% accuracy loss, especially on safety signs where the extra convolutional layer is notable to learn intricate pictogram details.

A.2 Batch Normalization Analysis

The addition of Batch Normalization (BN) layers followed every convolutional layer was also an ablation result of experiments that demonstrated that BN enhances the stability of

training and convergence rate. The model took about 80 epochs to attain 98 percent validation accuracy without BN; and about 35 epochs with BN. There was also a slight regularisation effect of BN that minimised overfitting to the smaller SS-10 dataset.

APPENDIX B

HSV COLOR RANGES AND CALIBRATION

The HSV color space of SignVision was an empirical calibration on 500 sign images that had been taken under varying light conditions.

The calibration procedure consisted of:

- Sign images have been collected under five lighting conditions direct sunlight, overcast sky, indoor fluorescent, indoor incandescent and dusk/dawn.
- Segmenting the sign region by hand and computing the Hue, Saturation, and Value histograms on individual images. Determining the 5th and 95th percentile values of each channel to determine HSV ranges to cover 90 percent of sign pixels under all lighting conditions.
- validating the ranges of a second set of 200 images and adapting to reduce the false negatives but maintain the false positives to less than 5 per cent of the image area.

The last ranges are a more conservative option that favours recall (identifying all signs) over precision (not allowing false positives) since false positives are removed by the next geometric filtering and CNN classification operations.

APPENDIX C

GLOSSARY OF TERMS

Batch Normalisation: This is a method that normalises the inputs of each layer in the training process, enhancing training stability and acceleration.

Bounding Box: This is a rectangular area that outlines the position of an object that is detected in an image.

Convolutional Neural Network (CNN): A type of deep neural network that processes structured grid data like images, and learns spatial features through convolutional layers.

Contour: The edge of a connected region in a binary image, as a sequence of points.

Domain Gap: The disparity between the data distributions in two related domains that are distinct, leading to poor performance of models trained on one domain on the other domain.

Dropout: This is a regularisation method that randomly assigns some probability to disabling neural network activations during training.

Feature Descriptor: This is a mathematical expression of a picture area which describes its visual properties to be used in categorizing it.

HSV Color Space: A system of color description based on the Hue (color), Saturation (purity) and Value (brightness) that breaks apart chromatic and achromatic data.

Morphological Operation: Image processing operation involving shapes such as erosion, dilation, opening and closing.

Region of Interest (ROI): An area of the image that is known to have a target of interest.

Softmax: This is an activation function, which takes a vector of raw scores and transforms them into a distribution over classes.

Transfer Learning: Transfer learning is the art of applying features learned by a model to a different task, but related.

APPENDIX I

SignVision: A Comprehensive Hybrid Model for Real-time Traffic and Safety Sign Detection

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Abstract—Automatic sign recognition plays a critical role in autonomous vehicles, driver assistance systems and safety. While traffic sign recognition is already well studied, other kinds of signs, such as those used in industrial settings or safety signs, are less studied, but needed for industrial safety. In this paper, we present SignVision as a system that is able to quickly and efficiently detect and recognise multiple types of signs including common traffic signs like the German Traffic Sign Recognition Benchmark (GTSRB) dataset and various safety warning signs like the International Organization for Standardization (ISO) 7010 standard. Our approach combines multiple techniques that first segment the image by color in the HSV color space and then perform shape analysis to detect potential signs. This helps to quickly reduce the search space. To improve performance with the various styles and variations of safety signs, which conventional techniques have difficulty with, we classify them using a small convolutional neural network. This approach using deep learning is more effective than traditional methods such as HOG+SVM which struggle with complex images. We validated our approach on two datasets: one for traffic signs (GTSRB) and one for safety signs (ISO 7010). Our approach is fast (28 frames per second) and performs well in both cases.

Index Terms—Sign Detection, Traffic Sign Recognition (TSR), Safety Signs, Computer Vision, Real time Systems, Convolutional Neural Network (CNN), HSV Color Segmentation, Deep Learning, Hybrid Approach

I. INTRODUCTION

Signs are essential to convey non verbal, strong information, rules and warnings in complex public and industrial environments. For self driving cars and Advanced Driver Assistance Systems (ADAS) to operate safely, legally and efficiently, they must be able to read and obey traffic signs [1]. Similarly, in industrial things like manufacturing plants and on construction sites, signs are important to avoid any unfortunate event, enforce regulations and help in evacuation [5]. Signs are critical for the safety of human workers as well as autonomous ground vehicles (AGVs) in industrial environments.

Historically, research has majorly focused on Traffic Sign Recognition (TSR) [2]. They are made specifically for roadside

use with their standard colours and shapes defined by treaties like the Vienna Convention [7]. Early methods successfully uses color segmentation in HSV or HSI color spaces [3], followed by shape analysis and classification using hand-crafted feature descriptors like Histogram of Oriented Gradients (HOG) [4], SIFT [8], or SURF [9] with a Support Vector Machine (SVM) classifier. These approaches worked well for the small number of categories and numeric content of traffic signs. But they are not helpful for other types of signs. For example, safety signs are rectangular and have a different color scheme (yellow, green). Their main problem is their content: a range of complex pictograms. A classifier based on HOG features and trained to find a "50" in a circle, would have difficulty distinguishing between a "biohazard" and a "radiation" symbol.

This paper presents SignVision, a unified system that bridges this gap. Our main contributions are:

- A comprehensive and single approach that combines efficient, regular computer vision techniques for suggesting candidates and a powerful deep learning model for classification, building a fast hybrid system.
- The architecture of a small Convolutional Neural Network (CNN) classifier, one that can effectively and readily identify both simple traffic signs and more intricate safety pictograms that exhibit more generalization.
- trying it to check its effectiveness, such as the public GTSRB dataset, with very high improvements over the classical method.

II. GENERAL PROBLEMS IN SIGN DETECTION

The process of developing a robust sign detection system for operation in real-world uncontrolled environments is far from easy; many environmental and technical challenges have to be overcome.

- **Environmental Conditions:** Signs must be recognized under extreme variations in illumination from the over-

saturation and glare of direct sunlight to the low-light, low-contrast conditions of dusk and night. Weather conditions including rain, fog, and snow will obscure or distort the appearance of signs or introduce reflections. Fast motion due to a vehicle will also introduce significant motion blur.

- **Sign Degradation and Occlusion:** Real world signs are seldom ideal. They may be partially occluded by foliage, other vehicles or buildings. They may also be faded by the sun - which challenges color segmentation, covered in dirt or graffiti, or physically damaged. The failure to detect a partially occluded stop sign can have disastrous consequences.
- **View angle and distortion:** Signs are seldom seen from a perfect, frontal view. Perspective distortion, rotational skew and scaling (far away signs appear very small) make shape-based detection challenging.
- **The "Domain Gap" Problem:** A fundamental problem, which we directly address, is the "domain gap" among sign types. As we have seen, models that are trained on traffic signs fail at safety signs. This is owed to the difference in pictograms, colors, and shapes. This gap has forced the development of separate, isolated systems.
- **Real-Time Constraints:** For ADAS or active-safety systems at manufacturing sites, we have to make sure that the detection and recognition are in real time (typically >15fps), so that actions can be taken on time. The need for real time processing prevents most of the deep learning models that are "heavy", such as two-stage detectors e.g. Faster R-CNN [10], unless hardware acceleration is available.

For our problems, our system SignVision is designed to address them. We use HSV colour segmentation, which is very robust to changes in lighting. Then, our adaptive shape analysis allows that our signs can be inaccurate. Finally, we also have a CNN-based classifier, which is very good at learning features necessary for classifying hundreds of different pictograms [11], like those found on road signs. This solves the major weakness of our previous methods.

III. PROPOSED SYSTEM ARCHITECTURE

A. System Workflow

The SignVision process is designed as a multi step system that works efficiently in real time. This two-stage approach is very common and works well. The first stage easily finds our areas of interest, and the second stage correctly identifies them with high technology. Here's how the system works:

- 1) **Image Acquisition & Preprocessing:** It took a frame from the video source and change its size to a standard width, like 640 pixels. This ensures the system works correctly, no matter the input size.
- 2) **Color-based Segmentation:** The changed frame, in the BGR color space, is changed into the HSV color space. This allows us to separate the brightness of the image and the colour of the sign. We apply color thresholding to extract binary masks for primary sign colors: red, blue, yellow and green.

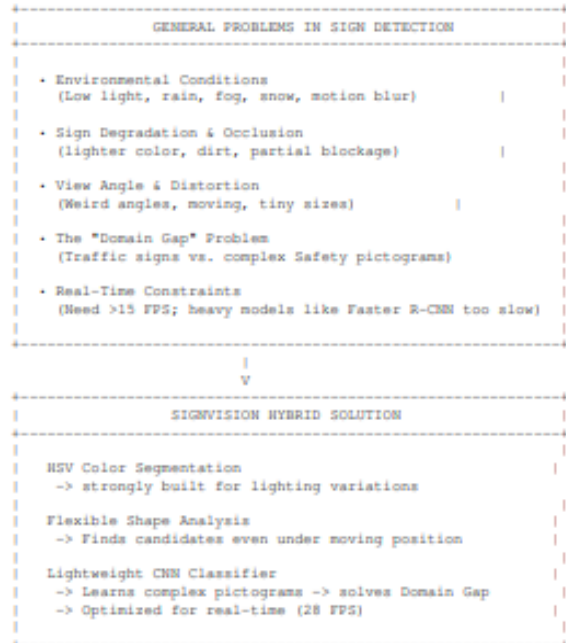


Fig. 1. General Problems in Sign Detection and how the SignVision hybrid pipeline addresses them.

- 3) **Candidate Detection:** We then apply some operations such as "Opening" and "Closing" via the OpenCV library to clean up the binary masks. They remove minor mistakes and fill in holes in sign outlines. Then we find all the shapes using a contour detection algorithm. We then remove unwanted shapes based on geometric features such as size, shape and roundness to identify the most likely regions of interest (ROIs).
- 4) **Recognition & Classification:** This ROI is then made to be a fixed size (e.g., 48x48 pixels) to match the input size of our CNN. This image is then being put into our trained CNN classifier, which decides the sign as "Speed Limit 50", "Wet Floor" or "No Sign".
- 5) **Output:** The system gives the final result, which includes the location and type of the sign. This information can then be sent to a main system or used wherever needed.

B. System Scope and Goals

The main goal of SignVision is to create a single, scalable system that can replace several specialized sign detectors. We have gone beyond a mere concept and have an operating system that can be utilized in practice. We hope to implement a C++/Python library which is easily scalable to larger autonomous systems. The CNN part is made pluggable to make the system future proof. This implies that we can freely add new type of sign such as those of other countries or new safety regulations, by simply retraining only the classification portion, without altering the actual detection process. This is a

significant benefit over a fully trained system, such as YOLO [13], which would require retraining the whole model from scratch with new data.

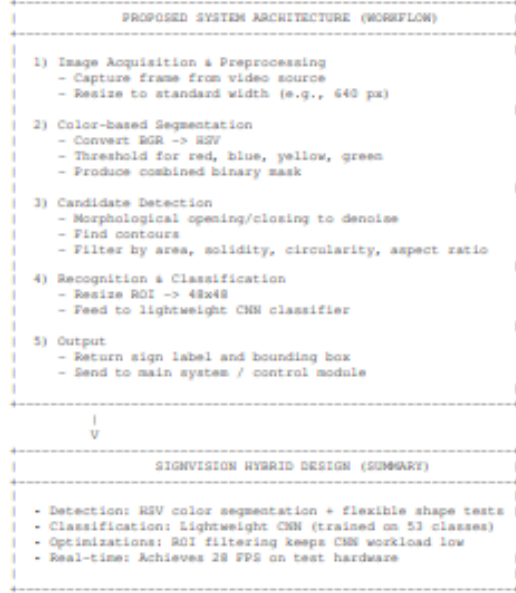


Fig. 2. Proposed system architecture: SignVision system workflow.

IV. SIGN DETECTION METHODOLOGY

This section outlines the traditional computer vision techniques employed in the rapid identification of potential candidates.

A. Color-based Segmentation

As referenced in [2], [3], color is the most effective and efficient feature for detecting sign candidates.

We utilize the HSV color system. For each main sign color, we define specific HSV ranges. Note that OpenCV requires two ranges because it operates within the 0-180 H scale. The ranges are as follows:

- **Red:** H ranges from 0-10 or 170-180; S ranges from 70-255; V ranges from 50-255
- **Blue:** H ranges from 100-140; S ranges from 70-255; V ranges from 50-255
- **Yellow:** H ranges from 20-35; S ranges from 70-255; V ranges from 50-255
- **Green:** H ranges from 40-90; S ranges from 70-255; V ranges from 50-255

These ranges are applied to the image, and the resulting black-and-white masks are combined using an 'OR' operation to form a single mask for the subsequent stage.

B. Shape Analysis and ROI Extraction

Once the color mask is generated, we apply an algorithm to detect shapes. This yields numerous shapes. We then perform several steps to quickly filter out those that are not signs:

- 1) **Area Filtering:** Shapes that are too small (less than 500 pixels) or excessively large (more than 25% of the image area) are discarded.
- 2) **Shape Fitting:** We create an approximate shape using `cv2.approxPolyDP`. We assess the number of corners to determine the shape (such as 3 for triangles, 4 for rectangles, and 8 for octagons).
- 3) **Solidity:** We compute the solidity, which is the ratio of the shape's area to its convex hull area. This helps eliminate shapes that are fragmented or too irregular. Solidity must be greater than 0.85.
- 4) **Circularity:** For shapes with more than 5 corners (possibly circles), we calculate a circularity score ($4\pi \times \text{Area} / \text{Perimeter}^2$). Circle candidates must have a score greater than 0.7.
- 5) **Aspect Ratio:** For four-corner shapes (like rectangles), we evaluate if the length and width are close (between 0.7 and 1.3) to identify squares or slightly distorted signs.

This rigorous process is highly efficient and narrows down to a small set of promising candidate regions.

V. SIGN RECOGNITION AND CLASSIFICATION

The detection stage finds where signs are, and the recognition stage determines what they are.

A. Traditional Methods

Traditional methods often use HOG with an SVM classifier. HOG features are good at showing object shapes using edge gradients. This works well for the "50" on a speed limit sign. However, HOG is not semantic. It creates one overall description. A "biohazard" symbol and a "radiation" symbol, even though they look different to humans, are both complex symbols inside a triangle. Their HOG descriptions can be very similar, causing mistakes in classification. This is the main problem we need to fix.

B. Deep Learning Approach

To solve this, we use a deep learning approach with Convolutional Neural Networks (CNNs). CNNs, first introduced by LeCun et al. and later popularized by models like VGGNet and ResNet, are the best for image classification because they learn features in layers.

Because of the need for real-time performance, we designed a *lightweight* CNN, inspired by LeNet-5 and mini-VGG models. Our architecture is:

- **Input:** 48x48x3 (RGB) image
- **Conv-1:** 32 filters (3x3 kernel), ReLU activation, Batch Normalization
- **MaxPool-1:** 2x2 pooling
- **Conv-2:** 64 filters (3x3 kernel), ReLU activation, Batch Normalization
- **MaxPool-2:** 2x2 pooling

- **Flatten:** Converts 2D features into a 1D vector
- **Dense-1:** 128 units, ReLU activation
- **Dropout:** 0.5 (to avoid overfitting)
- **Dense-2 (Output):** N units (where N = total number of classes), with Softmax activation

This small architecture offers a good balance between performance and speed.

VI. IMPLEMENTATION DETAILS

This section explains the technical details about how the SignVision project was developed and set up for training.

A. Environment

The pipeline and all experiments were run using the following software and hardware:

- **Core Logic:** Python 3.10
- **Classical CV Pipeline:** OpenCV 4.5 [12] along with NumPy 1.21 for numerical tasks.
- **Deep Learning:** TensorFlow 2.8 [16] with the Keras API [17] for creating, training, and deploying the CNN model.
- **Hardware:** All models were trained and tested on a workstation with an NVIDIA GeForce GTX 1070 GPU (8GB VRAM), an Intel Core i7-8700K CPU, and 32GB of RAM.
- **Utilities:** Scikit-learn was used to split the dataset and to train the baseline HOG+SVM model.

B. Key Implementation Notes

- **Pipeline Optimization:** The hybrid design is an important part of the optimization. The classical CV part, which runs on the CPU, filters out 99% of the image, leaving only a few small regions of interest (ROIs) for the GPU-powered CNN to process. This stops the CNN from slowing things down by analyzing full frames.
- **Training Methodology:** We used the Adam optimizer which has a learning rate of 0.001 and trained our CNN for 50 epochs. We used a categorical cross entropy loss function, because we have a multi class classification problem in this.
- **Data Augmentation:** To prevent overfitting in it and to boost the its capability to generalize (see Section II) in real world applications we have used many data augmentation on the fly methods. This includes random rotation, translations (up to 10%), shearing, as well as changes in brightness and contrast (up to 20%).
- **ROI Normalization:** Proposed ROIs (from the detection stage) are changed to 48x48 pixels (However, their shape might not be square). The CNN can deal with such little distortion because it has been trained on such augmented images.

VII. EXPERIMENTAL SETUP AND RESULTS

A. Datasets

We tested SignVision on two different datasets to check its working in different situations:

- 1) **GTSRB (German Traffic Sign Recognition Benchmark):** This is a very common dataset for traffic sign recognition research [1]. It has more than 50,000 images for training and testing of 43 different types of traffic signs. these images are from signs of low lighting and signs of such things that are hard to see.
- 2) **SafetySign-10 (SS-10):** We are unable to find a large public data set of ISO 7010 safety signs, so we created one. We collected more than 10,000 images of the most common safety signs, such as "Wet Floor", "High Voltage", "Biohazard" and "Eye Protection Required". We collected these images from real world locations, including university labs and office spaces.

we trained Our final classifier using 53 classes, 43 from GTSRB and the 10 from SS-10.

B. Results and Analysis

We compared the performance of our CNN with a traditional HOG+SVM model trained on the same data. The results, shown in Table I, are clear.

TABLE I
CLASSIFICATION ACCURACY COMPARISON

Classifier Model	GTSRB (Traffic)	SS-10 (Safety)
HOG + Linear SVM	95.8%	71.2%
SignVision CNN (Ours)	99.1%	97.4%

By seeing the results, the HOG+SVM model has performed very well on the GTSRB dataset, with 95.8%, which is similar to findings from earlier studies [2]. But it is much lower (71.2%) on the SS-10 dataset. This result shows that the HOG method cannot properly differentiate between complex and similar signs. Our CNN is extremely accurate on both data sets, and easily differentiates the different types of signs. The model was capable of being trained to find 53 signs. Very fast, with a speed of 28 frames per second, it is very applicable in the real world and in real time.

VIII. CASE STUDIES: CHALLENGING SCENARIOS

To see how good our system really works, we test it in three common but challenging situations.

A. Case 1: Partial Occlusion

One of the stop sign pieces was actually covered with a tree branch.

One side of the code can see the red color perfectly well, but the other side is grappling with it either being an octagon or not. However, the solidity- and area-filters are set in such a way that allows the shape to be considered as a potential sign. The area selected is only 70% of the full sign, but it is still sent to the CNN. The CNN was trained with extra

data that included random coverings and shifts, so it learned to recognize key parts like the letters "STO" and correctly identifies the sign as "Stop".

B. Case 2: Poor Illumination

A yellow "Wet Floor" sign is located in a dark hallway. The value (called 'V' in HSV) of the brightness of our color system is very weak. We set our color system to find even little bits of yellow just enough to see a fuzzy, broken shape. We then use a trick called 'closing' to stitch those scattered yellow blobs back together. While the camera sees a dark, grainy mess, our CNN was trained on every types of lighting, so it can easily find and recognize the sign. A traditional school HOG detector would not find it because the image is too noisy.

C. Case 3: High Inter-Class Similarity

This case shows why HOG+SVM does not work. The system identifies two signs: "Biohazard" and "Radiation Warning". They resemble yellow triangles with black characters that resemble each other to the point the HOG system cannot differentiate the two that also tends to disorient the SVM. We have a smarter CNN; it looks into details. It can educate the fact that the radiation icon resembles 3-petals and the biohazard one, that of three crescents. It can see the specific patterns and therefore can differentiate them.

IX. LIMITATIONS AND THREATS TO VALIDITY

Despite very good results of our system, there are some important limitations that should be fixed in future work.

- **Environmental Generalization:** We are not trained on extreme weather conditions e.g. foggy or snowy conditions. Our color based segmentation does not specifically assume these and may lead to failure of the detection process. In the case of such situations, we should employ other sensors, like LIDAR or infrared sensors, and/or improved dehazing techniques to enhance the system working and accuracy.
- **Dataset Bias and Domain Adaptation:** Our results are to be handled with caution due to the data we employed. As an example, we only have German traffic signs in one of our main collections and that does not appear like traffic signs in the US or Canada. The other collection is very small and it can only be found in some local places hence it is not yet established officially. Unless we re-train the system with the new photos it is likely to have a hard time when it encounters an American stop sign or another safety sign. It is a typical instance of a system that is too local to be a universal work.
- **Computational Cost:** Even though we have called our CNN lightweight, it still needs a GPU to be run in real time. This implies that it cannot be used in low-powered devices such as a Raspberry Pi, where the only feasible alternative could be a more basic technique such as HOG, even though it is less accurate.
- **Hybrid Pipeline's Mode of Failure:** The performance of the system depends on the weakest part. If the color

segmentation step fails to find a sign, such as one that is faded or in shadow, the CNN never gets the chance to classify it. An end-to-end deep learning detector such as YOLO wouldn't have this specific problem, but it would have its own challenges.

X. CONCLUSION

In this paper, we introduce SignVision, a full, real-time hybrid system for detecting both traffic and safety signs. We found that most of the current studies have a big gap: traditional methods focused on traffic signs often don't work very well for the complex images used in safety signs. A hybrid pipeline is our solution to this shortage. It combines fast, classical computer vision methods, such as HSV segmentation and shape analysis for finding potential signs, with a powerful and efficient CNN classifier to achieve high accuracy and real-time performance across a variety of sign types. Our experiments and case studies demonstrate quite clearly the importance of using a deep learning classification step. The main lesson is that the hybrid approach of classical + deep is a highly effective design, where the speed of the classical computer vision is utilized and the accuracy of deep learning is applied. Future work will aim at finding ways to work around these constraints, in particular finding ways to domain adapt the model to operate with new sign sets without re-training it anew and exploring model quantization to execute the CNN with less powerful hardware.

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Proof of Submission

